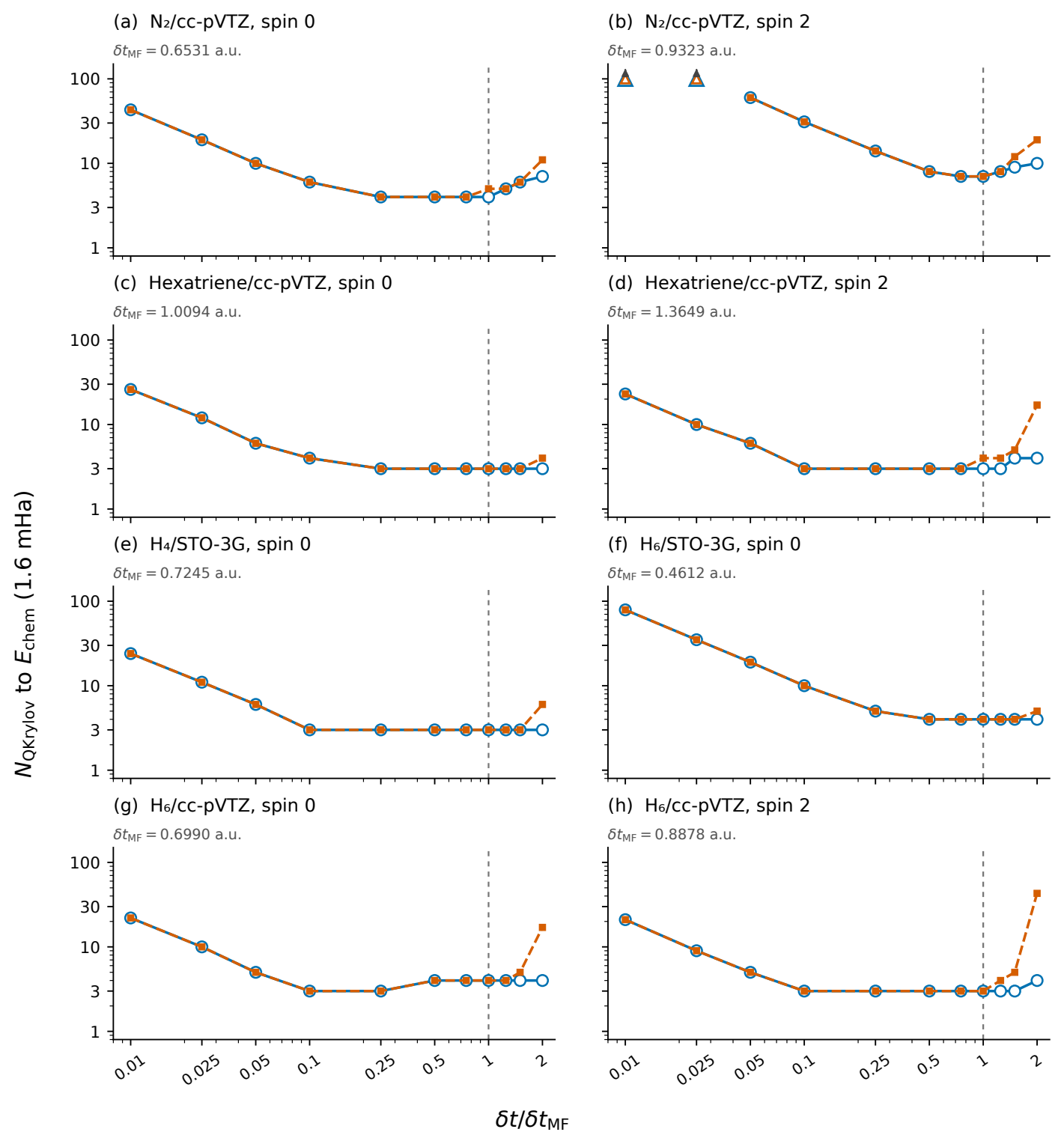

(a) N₂/cc-pVTZ, spin 0
δt_MF = 0.6531 a.u.
(b) N₂/cc-pVTZ, spin 2
δt_MF = 0.9323 a.u.
(c) Hexatriene/cc-pVTZ, spin 0
δt_MF = 1.0094 a.u.
(d) Hexatriene/cc-pVTZ, spin 2
δt_MF = 1.3649 a.u.
(e) H₄/STO-3G, spin 0
δt_MF = 0.7245 a.u.
(f) H₆/STO-3G, spin 0
δt_MF = 0.4612 a.u.
(g) H₆/cc-pVTZ, spin 0
δt_MF = 0.6990 a.u.
(h) H₆/cc-pVTZ, spin 2
δt_MF = 0.8878 a.u.
N_QKrylov to E_chem (1.6 mHa)
100
30
10
3
1
0.01
0.025
0.05
0.1
0.25
0.5
1
2
δt/δt_MF
Exact exponential QRTE
Second-order Trotter QRTE (one slice)
δt = δt_MF
No hit within 100 iterations